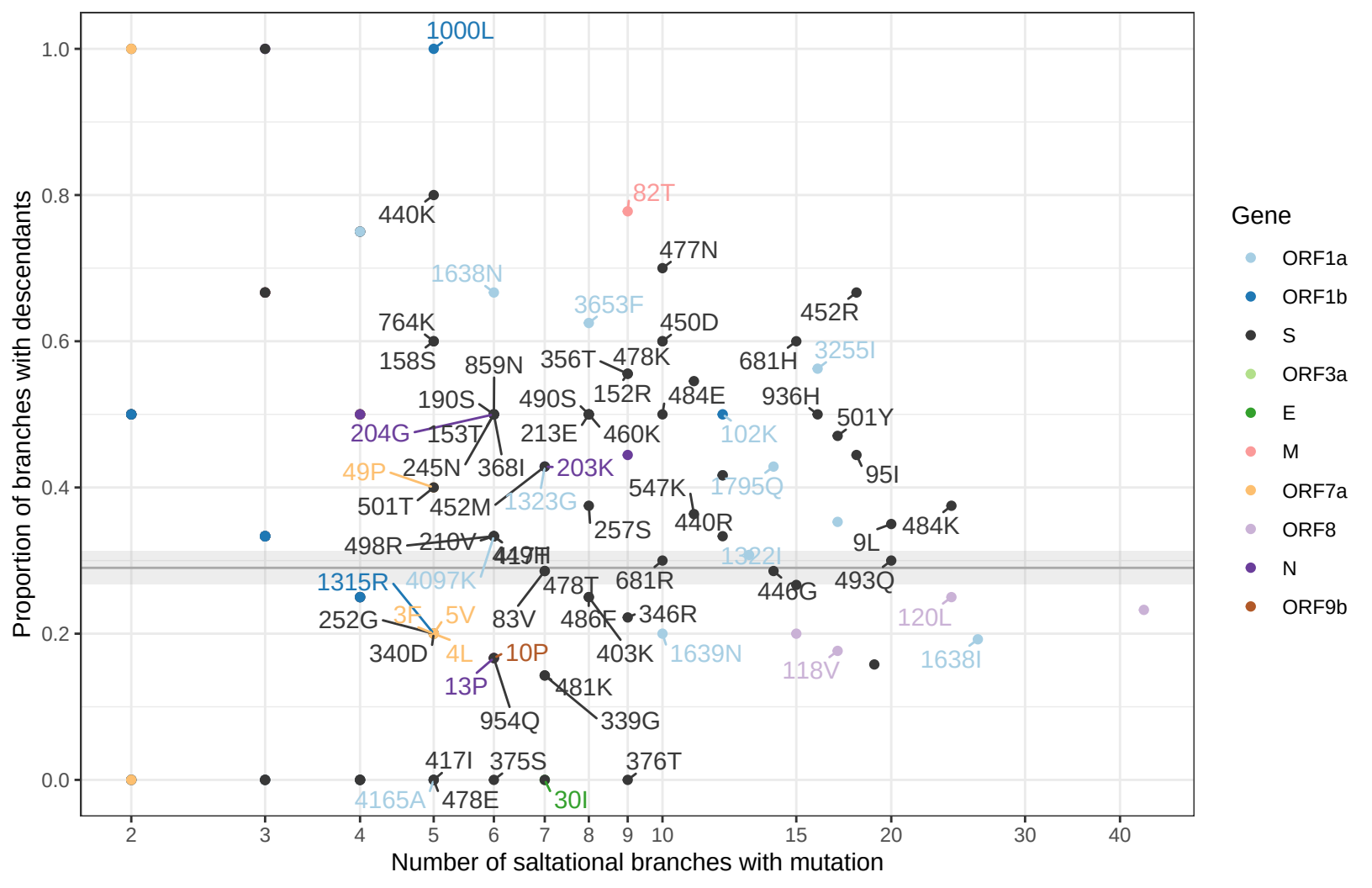
Proportion of branches with descendants
1.0
0.8
0.6
0.4
0.2
0.0
Number of saltational branches with mutation
2
3
4
5
6
7
8
9
10
15
20
30
40
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF7a
ORF8
N
ORF9b
1000L
440K
82T
477N
1638N
3653F
452R
764K
450D
158S
859N
356T
478K
681H
3255I
190S
490S
152R
484E
936H
501Y
204G
153T
213E
460K
102K
49P
245N
368I
203K
547K
1795Q
95I
501T
452M
1323G
257S
440R
484K
498R
210V
9L
1315R
4097K
478T
681R
493Q
252G
3F
5V
83V
486F
346R
120L
340D
4L
10P
403K
1639N
118V
1638I
13P
481K
954Q
339G
417I
375S
376T
4165A
478E
30I